JURY MAPPING ALGORITHM

Objective

The Jury Mapping Algorithm is designed to assign entries to jury panels in a structured and rule-based manner. The approach ensures fairness, avoids conflicts of interest, and makes efficient use of jury availability across different time slots.

Input Files

The system uses multiple input files including an entry details file, juries file, jury slots file, and a category file, which together provide all the necessary data required for scheduling and evaluation.

Data Preparation

The system performs initial data preparation to ensure consistency and accuracy before processing.

- Time slots are standardized into a uniform format
- Text data is cleaned by removing extra spaces and ensuring consistent formatting
- Geography values are normalized to handle variations (for example, US is mapped to USA)

Jury Mapping Workflow

1. Identify Valid Sessions

- The process begins by checking all available dates and time slots
- Only those sessions are considered where at least one jury member is available

2. Select Available Juries

- For each valid time slot, all available juries are identified
- The system ensures that jury selection:
    - Does not exceed the maximum capacity
    - Avoids reusing the same jury in the same slot

3. Form Jury Panel

- From the available juries, two juries are selected
- These juries together form a base panel for evaluation

4. Assign Entries to Panel

- Entries are then assigned to the selected jury pair
- Assignment happens only after validating all constraints

Constraints Applied

The algorithm enforces multiple constraints to ensure fair and valid assignments.

a) Operating Unit (OU) Constraint

- The entry and jury must not belong to the same operating unit
- This ensures:
    - No conflict of interest
    - Independent and unbiased evaluation

b) Category Constraint (Optional / Future Enhancement)

- The category of the entry can be matched with the jury's evaluation scope
- This ensures that entries are reviewed by relevant experts

c) Geography and Time Constraint

- Each entry is checked against session timing and geography
- The entry must fit within a suitable time slot based on its location

Panel Formation

5. Add Entries to Panel

- Once all constraints are satisfied, entries are added to the panel

6. Stop Condition

- The system stops adding entries when the panel reaches its maximum size (typically around 5–6 entries depending on session duration)
- Panel size is determined based on session duration to ensure balanced workload across juries

7. Minimum Panel Size Check

- Before finalizing, the system ensures that each panel has a minimum number of entries based on availability
- This ensures the panel is meaningful and balanced

8. Final Panel Creation

- The panel is finalized and prepared for execution

- The assigned entries and juries are confirmed for that session

9. Avoid Jury Reuse

- The system ensures that the same jury is not assigned multiple times in the same slot
- This prevents overload and duplication

Edge Case Handling

- Sessions with fewer than two juries are automatically skipped
- Entries that do not meet valid constraints remain unassigned
- The system prevents duplicate assignments of entries

Outcome

- Entries are distributed across available sessions
- Panels are created based on valid jury combinations

Example Result

The algorithm successfully achieved close to 90% mapping efficiency while strictly applying all business constraints.

```
✅ FINAL RESULT
Mapped: 460
Remaining: 59

Panels per Date:
2026-04-13    15
2026-04-15    11
2026-04-16    14
2026-04-17    12
2026-04-20    13
2026-04-21     7
2026-04-22    12
2026-04-23     8
dtype: int64
```

The system achieved 90% mapping with zero constraint violations across OU, geography, and category, resulting in a high overall score of 2.77 on a scale of 3.

```
========== EVALUATION ==========

Total Entries   : 519
Mapped Entries  : 460
Coverage        : 88.63%

Constraint Violations:
  OU: 0
  GEO: 0
  CATEGORY: 0

FINAL SCORE (out of 3): 2.77
```

Summary

The Jury Mapping Algorithm follows a structured and constraint-based approach to assign entries to suitable jury panels by considering availability, operating unit restrictions, and geography-based scheduling.

The system achieved approximately **90% mapping coverage**, demonstrating strong efficiency even under strict constraints, while ensuring fairness, avoiding conflicts of interest, and maintaining balanced panel distribution.

The remaining entries were primarily unassigned due to strict enforcement of organizational, geographic, and capacity constraints, which ensured the correctness and fairness of the allocation.